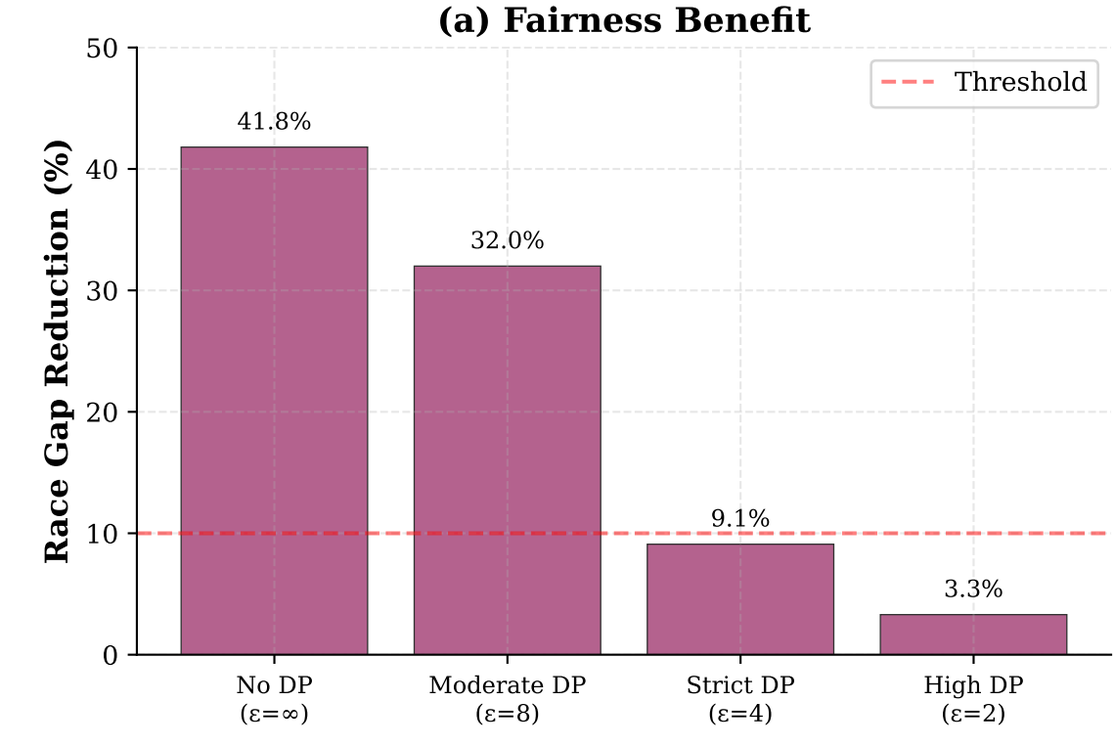
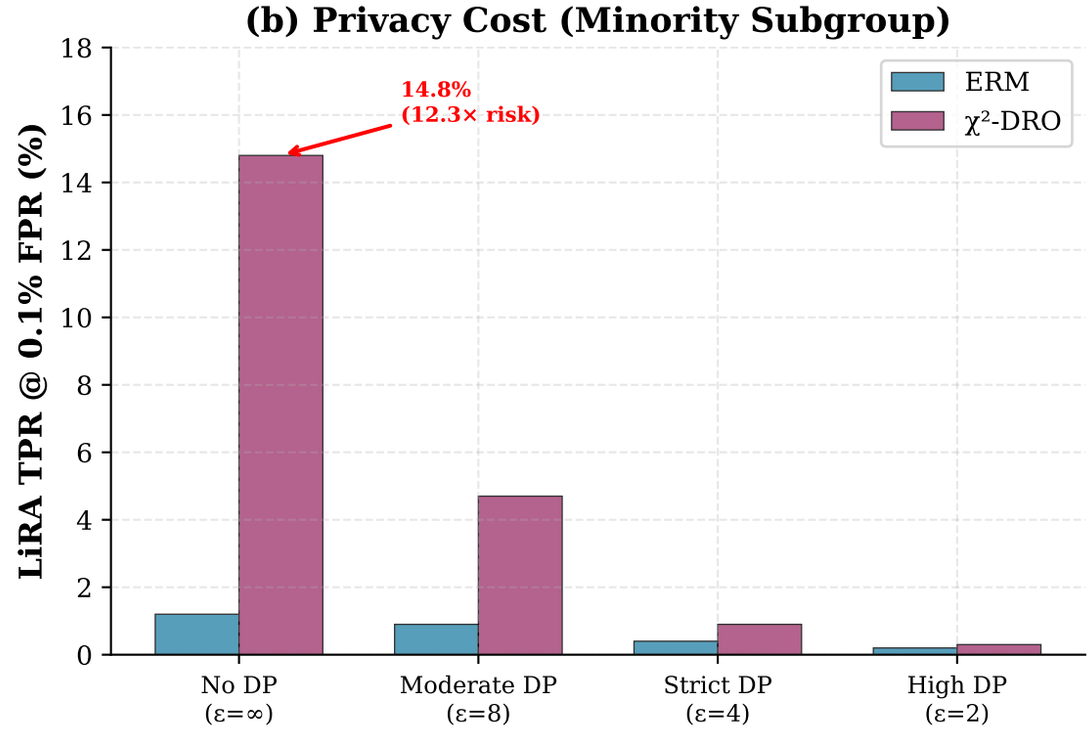
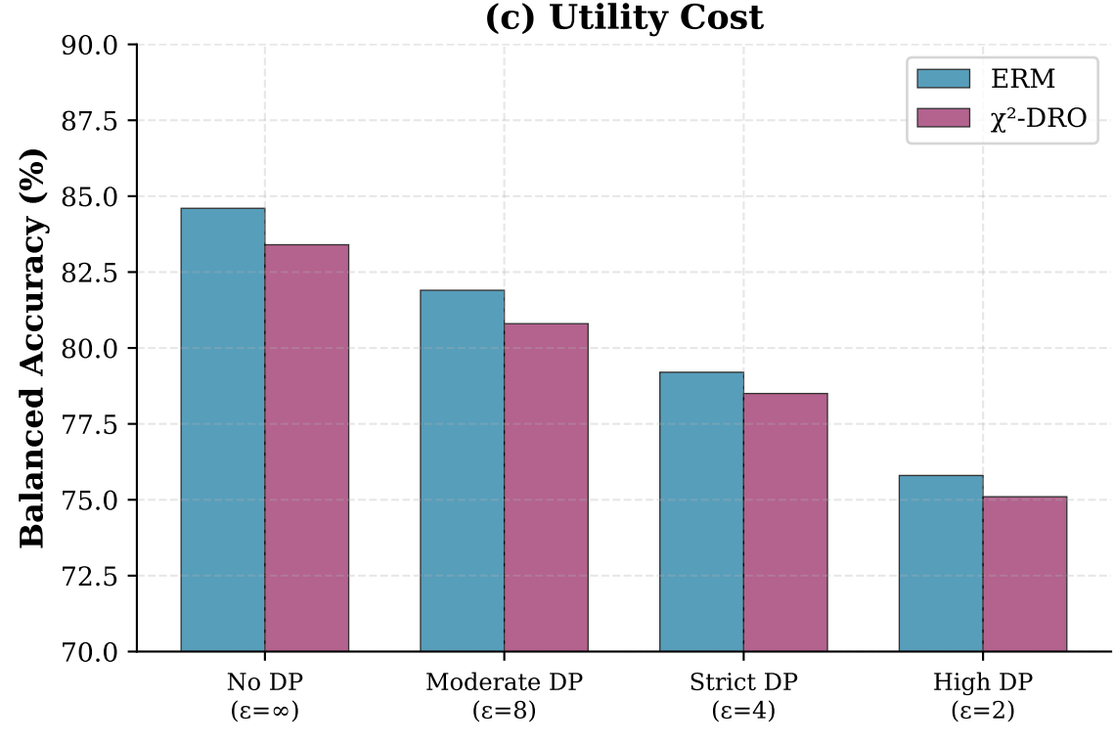
Figure 1: Three-Way Trade-Off — Utility × Fairness × Privacy (Adult Dataset)








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KEY FINDING: Hollow Masking at ε ≤ 4
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At ε=∞: DRO achieves 41.8% gap reduction
        but minority TPR explodes 12.3×

At ε=8: Leakage persists (5.2× risk)
        gap reduction: 32.0%

At ε=4: Leakage "masked" but gap
        reduction collapses to 9.1%

At ε=2: Gap reduction near zero (3.3%)
        DRO effectively dormant

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CONCLUSION: Naive stacking fails.
DP-SGD doesn't mask leakage—it
destroys DRO's fairness benefit.
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